18/3/20

Unit-4.

Spectral Densities And Linear Systems with Random inputs :~

power spectral density (or) Power density spectrum.

def: The power spectral density $S_x(\omega)$ of a continuous time random process $x(t)$ is defined as the fourier transform of $R_{xx}(\tau)$.

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-i\omega\tau}\, d\tau$$

$$\because F[f(x)] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-isx}\, dx.$$

Properties:

1). The power spectral density fn of a real random process is an even function.

proof:

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-j\omega\tau}\, d\tau$$

$$S_{xx}(-\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{j\omega\tau}\, d\tau.$$

Let, $\tau = -u$, $\tau = -\infty \rightarrow u = \infty$.

$d\tau = -du$ $\quad$ $\tau = \infty \rightarrow u = -\infty$.

$$S_{xx}(-\omega) = \int_{\infty}^{-\infty} R_{xx}(-u)\, e^{-i\omega u}\, (-du).$$

$(\because R_{xx}(-u) = R_{xx}(u)$ is even$)$

$$= \int_{-\infty}^{\infty} R_{xx}(u)\, e^{-i\omega u}\, du.$$

$$-\int_{\infty}^{-\infty} = \int_{-\infty}^{\infty}$$

$$S_{xx}(-\omega) = S_{xx}(\omega).$$

$\therefore$ It is even fn

Q. The spectral density fn of a process $x(t)$ real (or) complex is real function of ω and non-negative.

Proof: $$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-i\omega\tau}\, d\tau$$

take conjugate on b/s.

$$S_{xx}^{*}(\omega) = \int_{-\infty}^{\infty} R_{xx}^{*}(\tau)\, e^{i\omega\tau}\, d\tau$$

$$= \int_{-\infty}^{\infty} R_{xx}(-\tau)\, e^{i\omega\tau}\, d\tau$$

Let, $u = -\tau$ $\quad [\because R_{xx}^{*}(\tau) = R_{xx}(-\tau)]$

$du = -d\tau$ $\quad t = -\infty \rightarrow u = +\infty$

$\tau = \infty \rightarrow u = -\infty.$

$$= -\int_{\infty}^{-\infty} R_{xx}(u)\, e^{-i\omega u}\, du.$$

$$S_{xx}^{*}(\omega) = \int_{-\infty}^{\infty} R_{xx}(u)\, e^{-i\omega u}\, du.$$

$$S_{xx}^{*}(\omega) = S_{xx}(\omega)$$

$\therefore$ It is real fn

Wiener - Khinchine Theorems :

If $X_T(\omega)$ is the fourier transform of the truncated random process defined as

$$x_T(t) = \begin{cases} x(t), & \text{for } |t| \le T \\ 0, & \text{for } |t| > T \end{cases}$$

where $\{x(t)\}$ is a real WSS process with power spectral density function $S(\omega)$, then.

$$S(\omega) = \lim_{T\to\infty} \left[\frac{1}{2T} E\{|X_T(\omega)|^2\} \right]$$

1. Find the power spectral density of a WSS process with Autocorrelation function.

i). $R_{xx}(\tau) = e^{-\alpha\tau^2}$

ii). $R_{xx}(\tau) = e^{-\tau^2/2}$

i).
Sol:

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-\alpha\tau^2} e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-(\alpha\tau^2 - i\omega\tau)}\, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-\alpha\left(\tau^2 + \frac{i\omega}{\alpha}\tau\right)}\, d\tau$$

$$\tau^2 + \underbrace{2\frac{i\omega}{\alpha}}_{b}\tau$$

$$\therefore \tau^2 + \frac{i\omega\tau}{\alpha} = \tau^2 + 2\tau\left(\frac{i\omega}{2\alpha}\right) + \left(\frac{i\omega}{2\alpha}\right)^2 - \left(\frac{i\omega}{2\alpha}\right)^2$$

$$\tau^2 + \frac{2i\omega}{\alpha}\tau + \left(\frac{i\omega}{2\alpha}\right)^2 - \left(\frac{i\omega}{2\alpha}\right)^2$$

$$= \left(\tau + \frac{i\omega}{2\alpha}\right)^2 - \left(\frac{i\omega}{2\alpha}\right)^2$$

$$= \int_{-\infty}^{\infty} e^{-\alpha\left[\left(\tau + \frac{i\omega}{2\alpha}\right)^2 + \frac{\omega^2}{4\alpha^2}\right]} d\tau.$$

$$= e^{\frac{-\alpha\omega^2}{4\alpha^2}} \int_{-\infty}^{\infty} e^{-\alpha\left(\tau + \frac{i\omega}{2\alpha}\right)^2} d\tau.$$

Let, $x = \sqrt{\alpha}\left(\tau + \frac{i\omega}{2\alpha}\right)$

$\therefore dx = \sqrt{\alpha}\, d\tau.$

$\frac{dx}{\sqrt{\alpha}} = d\tau.$

$$= e^{\frac{-\omega^2}{4\alpha}} \int_{-\infty}^{\infty} e^{-x^2} \frac{dx}{\sqrt{\alpha}}.$$

$$= e^{-\frac{\omega^2}{4\alpha}} \cdot \frac{\sqrt{\pi}}{\sqrt{\alpha}}.$$

$$= \sqrt{\frac{\pi}{\alpha}} \cdot e^{-\omega^2/4\alpha} \quad //.$$

Formula: ④. $\int_0^{\infty} e^{at} \cos bt\, dt = \frac{e^{at}}{a^2+b^2}(a\cos bt + b\sin bt).$

①. $\int_0^{\infty} x^n e^{-x}\, dx = n!$

②. $\int_0^{\infty} x^n e^{-ax}\, dx = \frac{n!}{a^{n+1}}$

③. $\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}.$

ii). **Sol:**

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} e^{-\tau^2/2} \cdot e^{-i\omega\tau} \, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\tau^2 + 2i\omega\tau)} \, d\tau.$$

$$\because (\tau^2 + 2i\omega\tau) = \tau^2 + 2i\omega\tau + (i\omega)^2 - (i\omega)^2.$$

$$= (\tau + i\omega)^2 + \omega^2.$$

$$= \int_{-\infty}^{\infty} e^{-\frac{1}{2}[(\tau+i\omega)^2 + \omega^2]} \, d\tau.$$

$$= e^{-\frac{\omega^2}{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(\tau+i\omega)^2} \, d\tau.$$

Let $x = (\tau + i\omega)\sqrt{1/2}$.

$\sqrt{2}\, dx = d\tau.$

$$= e^{-\omega^2} \int_{-\infty}^{\infty} e^{-x^2} \, dx \cdot \sqrt{2}$$

$$= e^{-\omega^2} \cdot \sqrt{\pi} \cdot \sqrt{2}$$

$$\boxed{= \sqrt{2\pi} \cdot e^{-\frac{\omega^2}{2}}.}$$

(5). The auto correlation function of the random telegraph signal process is given by $R(\tau) = a^2 e^{-2\delta|t|}$. Determine the power density spectrum of random telegraph signal.

Sol:-

Given, $R(\tau) = a^2 e^{-2\gamma|\tau|}$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} a^2 e^{-2\gamma|\tau|} \cdot e^{-i\omega\tau} \, d\tau.$$

$$= a^2 \int_{-\infty}^{\infty} e^{-2\gamma|\tau|} \cdot (\cos\omega\tau - i\sin\omega\tau) \, d\tau.$$

odd = 0.

$$= 2a^2 \int_{0}^{\infty} e^{-2\gamma|\tau|} \cos\omega\tau \, d\tau$$

$$= 2a^2 \left[\frac{e^{-2\gamma\tau}}{4\gamma^2+\omega^2} (-2\gamma\cos\omega\tau + \omega\sin\omega\tau) \right]_0^{\infty}$$

$$= 2a^2 \left[0 - \frac{1}{4\gamma^2+\omega^2} (-2\gamma) \right]$$

$$\boxed{= \frac{4a^2\gamma}{4\gamma^2+\omega^2}.}$$

Formula:

If $F[f(t)] = F(\omega)$ then.

i). $F[e^{-a|\tau|}] = \frac{2a}{e+\omega^2}$

ii). $F[\cos(\omega_0\tau)] = \pi[\delta(\omega-\omega_0) + \delta(\omega+\omega_0)]$

iii). $F[\sin(\omega_0\tau) = \frac{\pi}{i}[\delta(\omega-\omega_0) - \delta(\omega+\omega_0)]$

iv). $R_{xx}(\tau) = F^{-1}[S_{xx}(\omega)]$

v). $S_{xx}(\omega) = F[R_{xx}(\tau)]$.

vi). $R_{xx}(\tau - 2a) = e^{-i\omega 2a} = e^{-i2a\omega}$

vii). $R_{xx}(\tau + 2a) = e^{i\omega 2a} = e^{i2a\omega}$.

(5). A Random process $\{X(t)\}$ is given by $X(t) = A\cos pt + B\sin pt$, where A and B are independent RV such that $E(A) = E(B) = 0$ and $E(A^2) = E(B^2) = \sigma^2$. Find the power spectral density of a process.

Sol:-

W.K.T, If $X(t) = A\cos pt + B\sin pt$.

i) $E(A) = E(B) = 0$.

ii) $E(A^2) = E(B^2) = \sigma^2$ (or) K.

$$R_{xx}(\tau) = \sigma^2 \cos p\tau \quad (\text{or}) \quad = K\cos p(t_1 - t_2)$$

$$S_{xx}(\omega) = F[R_{xx}(\tau)]$$

$$= F[\sigma^2 \cos p\tau]$$

$$= \sigma^2 F[\cos p\tau]$$

$$= \sigma^2 [\pi\{\delta(\omega - p) + \delta(\omega + p)\}]$$

$$S_{xx}(\omega) = \pi\sigma^2 \{\delta(\omega - p) + \delta(\omega + p)\}.$$

(6). The auto correlation of the random binary transmission is given by $R_{xx}(\tau) = \begin{cases} 1 - \frac{|\tau|}{T}, & |\tau| \le T \\ 0, & |\tau| > T \end{cases}$

Find the power spectrum.

(5). The power spectral density of the random process whose auto correlation function of

$$R(\tau) = \begin{cases} 1-|\tau| & , |\tau| \le 1 \\ 0 & , \text{otherwise.} \end{cases}$$

(4). sol:

Given, $R(\tau) = \begin{cases} 1-\frac{|\tau|}{T} & , \text{for } |\tau| \le T \\ 0 & , \text{for } |\tau| > T \end{cases}$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau)\, e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-T}^{T} \left(1-\frac{|\tau|}{T}\right) (\cos\omega\tau - i\underbrace{\sin\omega\tau}_{\text{odd } 0})\, d\tau.$$

$$= 2\int_{0}^{T} \left(1-\frac{|\tau|}{T}\right) \cos\omega\tau\, d\tau.$$

$$= 2\left[\left(1-\frac{\tau}{T}\right)\frac{\sin\omega\tau}{\omega} - \left(-\frac{1}{T}\right)\left(-\frac{\cos\omega\tau}{\omega^2}\right)\right]_0^T.$$

$$= 2\left[\left(0-\frac{\cos\omega T}{T\omega^2}\right) - \left(0-\frac{1}{T\omega^2}\right)\right]$$

$$= 2\left[-\frac{\cos\omega T}{T\omega^2} + \frac{1}{T\omega^2}\right]$$

$$\boxed{S_{xx}(\omega) = \frac{2}{T\omega^2}\left[1-\cos\omega T\right].}$$

(6). Sol: Given, $R(\tau) = \begin{cases} 1-|\tau| & , |\tau| \le 1 \\ 0 & , \text{otherwise} \end{cases}$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} (1-|\tau|)\, e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-1}^{1} (1-|\tau|)\,(\cos\omega\tau - i\sin\omega\tau)\, d\tau$$

$$= 2\int_{0}^{1} (1-|\tau|)\cos\omega\tau\, d\tau.$$

$$= 2\left[(1-|\tau|)\frac{\sin\omega\tau}{\omega} - (-1)\left(-\frac{\cos\omega\tau}{\omega^2}\right)\right]_0^1.$$

$$= 2\left[\left(0-\frac{\cos\omega}{\omega^2}\right) - \left(0-\frac{1}{\omega^2}\right)\right]$$

$$\boxed{= \frac{2}{\omega^2}\left[1-\cos\omega\right]}$$

Note:

Given the power spectral density function $S(\omega)$ the autocorrelation function $R(\tau)$ is given by the Fourier Inverse transform of $S(\omega)$

$$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega)\, e^{i\tau\omega}\, d\omega.$$

⑥. If the power spectral density of a WSS process is given $S(\omega) = \begin{cases} \frac{b}{a}(a-|\omega|), & |\omega| \le a \\ 0, & |\omega| > a \end{cases}$

Find autocorrelation function of process.

Sol:- $$R(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S(\omega) e^{i\omega\tau} \, d\omega$$

Given,

$$S(\omega) = \frac{b}{a}(a-|\omega|), \quad |\omega| \le a.$$

$$R_{xx}(\tau) = \frac{1}{2\pi}\int_{-a}^{a} \frac{b}{a}(a-|\omega|) \cdot (\cos\omega\tau + i\,\cancel{\sin\omega\tau})\, d\omega \quad \text{0.}$$

$$= \frac{2b}{2\pi a}\int_{0}^{a} (a-\omega)\cos\omega\tau \, d\omega.$$

$$= \frac{b}{\pi a}\left[(a-\omega)\cdot\frac{\sin\omega\tau}{\tau} - (-1)\left(-\frac{\cos\omega\tau}{\tau^2}\right)\right]_0^a.$$

$$= \frac{b}{\pi a}\left[\left(0 - \frac{\cos a\tau}{\tau^2}\right) - \left(0 - \frac{1}{\tau^2}\right)\right]$$

$$R_{xx}(\tau) = \boxed{\frac{b}{\pi a \tau^2}\left[1-\cos a\tau\right]}.$$

$[\because \; 1-\cos 2\theta = 2\sin^2\theta$

$1-\cos\theta = 2\sin^2\frac{\theta}{2}.\;]$

$$R_{xx}(\tau) = \frac{b}{\pi a \tau^2} \cdot \left(2 \sin^2 \frac{a\tau}{2}\right)$$

$$\boxed{R_{xx}(\tau) = \frac{2b}{\pi a} \cdot \left[\frac{\sin\left(\frac{a\tau}{2}\right)}{\tau}\right]^2}$$

(7) Find the power spectral density random process $\{x(t)\}$, whose autocorrelation function is $R(\tau) = e^{-a\tau^2} \cos\omega_0\tau$.

Sol:- Given, $R(\tau) = e^{-a\tau^2} \cos\omega_0 t$.

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} R_{xx}(\tau) \cdot e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-a\tau^2} \cos\omega_0 t \cdot e^{-i\omega\tau}\, d\tau.$$

$$= \int_{-\infty}^{\infty} e^{-a\tau^2} \left(\frac{e^{i\omega_0\tau} + e^{-i\omega_0\tau}}{2}\right) e^{-i\omega\tau}\, d\tau.$$

$$= \frac{1}{2}\int_{-\infty}^{\infty} \left\{ e^{-a\tau^2 + i\omega_0\tau - i\omega\tau} + e^{-a\tau^2 - i\omega_0\tau - i\omega\tau} \right\} d\tau.$$

$$S_{xx}(\omega) = \frac{1}{2}\int_{-\infty}^{\infty} e^{-a\left[\tau^2 + \frac{i(\omega-\omega_0)}{a}\tau\right]} + e^{-a\left[\tau^2 + i\frac{(\omega_0+\omega)}{a}\tau\right]} d\tau \quad \hookrightarrow (1).$$

{W.K.T

$$R_{xx}(\tau) = e^{-a\tau^2} \longrightarrow S_{xx}(\omega) = \sqrt{\frac{\pi}{a}}\, e^{\frac{-\omega^2}{4a}}$$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} e^{-a\tau^2 - i\omega\tau}\, d\tau.$$

$$S_{xx}(\omega) = \int_{-\infty}^{\infty} e^{-a\left(\tau^2 + \frac{i\omega\tau}{a}\right)} d\tau.$$

$$= \sqrt{\frac{\pi}{a}}\, e^{-\omega^2/4a} \longrightarrow ②$$

compare ① & ②

$$S_{xx}(\omega) = \frac{1}{2}\left[\sqrt{\frac{\pi}{a}}\, e^{-\frac{(\omega_0+\omega)^2}{4a}} + \sqrt{\frac{\pi}{a}}\, e^{-\frac{(\omega_0+\omega)^2}{4a}}\right\}$$

$$S_{xx}(\omega) = \sqrt{\frac{\pi}{a}}\left(\frac{e^{-\frac{(-\omega_0+\omega)^2}{4a}} + e^{-\frac{(\omega_0+\omega)^2}{4a}}}{2}\right) //.$$

⑧. Given the power spectral density of random process is given by $S_{xx}(\omega) = \frac{\omega^2+9}{\omega^4+5\omega^2+4}$. Find the mean square value of the process.

Sol:- Let us take $u = \omega^2$.

$$S_{xx}(\omega) = \frac{u+9}{u^2+5u+4}.$$

$$S_{xx}(\omega) = \frac{(u+9)}{(u+4)(u+1)}$$

Apply partial fraction.

$$S_{xx}(\omega) = \frac{u+9}{(u+4)(u+1)} = \frac{A}{u+4} + \frac{B}{u+1}$$

$u+9 = A(u+1) + B(u+4)$ ..

put $u=-1$.

$8 = 3B$.

$B = 8/3$.

Put $u=-4$

$5 = -3A$

$A = -5/3$.

$$S_{xx}(\omega) = -\frac{5/3}{(u+4)} + \frac{8/3}{(u+1)}$$

$$= -\frac{5}{3} \cdot \frac{1}{(\omega^2+4)} + \frac{8}{3} \cdot \frac{1}{(\omega^2+1)}$$

$$= -\frac{5}{3} \cdot \frac{1}{(\omega^2+2^2)} + \frac{8}{3} \cdot \frac{1}{(\omega^2+1^2)}$$

Taking Inverse fourier.

$$= -\frac{5}{3} \cdot \left[\frac{1}{4} e^{-2|\tau|}\right] + \frac{8}{3}\left[\frac{1}{2} e^{-|\tau|}\right]$$

$$R_{xx}(\tau) = -\frac{5}{12} e^{-2|\tau|} + \frac{8}{6} e^{-|\tau|}$$

The mean square value, $\tau = 0$.

$$R_{xx}(0) = E[x^2(t)] = -\frac{5}{12} + \frac{8}{6}$$

$$= \frac{-5+16}{12} \boxed{= \frac{11}{12}.}$$

<u>Note</u>:

1. $F^{-1}\left[\frac{1}{a^2+\omega^2}\right] = \frac{1}{2a} e^{-a|\tau|}$

2. $F^{-1}\left[\frac{1}{(1+\omega^2)^2}\right] = \frac{1}{4}(1+\tau) e^{-\tau}$

3. Mean square value (or) Average value $= R_{xx}(0)$

$$E[x^2(t)] = R_{xx}(0)$$

(9) The power spectrum of a WSS process $\{X(t)\}$ is given by $S(\omega) = \frac{1}{(1+\omega^2)^2}$. Find its auto correlation function $R(\tau)$ and average power.

Sol:- Given, $S(\omega) = \frac{1}{(1+\omega^2)^2}$.

To find $R_{xx}(\tau)$:

$$R_{xx}(\tau) = F^{-1}[S(\omega)]$$

W.K.T,

$$F^{-1}\left[\frac{1}{(1+\omega^2)^2}\right] = \frac{1}{4}[1+\tau]\, e^{-\tau}$$

$$\boxed{\therefore R_{xx}(\tau) = \frac{1}{4}[1+\tau]\, e^{-\tau}}$$

To find average power:

$$R_{xx}(0) = \frac{1}{4}[1+0]\, e^{-0}$$

$$\boxed{= \frac{1}{4}.}$$

(10) The power spectral density function of a zero mean WSS process $\{X(t)\}$ is given by $S(\omega) = \begin{cases} 1, & |\omega| < \omega_0 \\ 0, & \text{elsewhere.} \end{cases}$ Find $R(\tau)$ and show that $X(t)$ and $X\left(t+\frac{\pi}{\omega_0}\right)$ are uncorrelated.

Sol:

$$R_{xx}(\tau) = \int_{-\omega_0}^{\omega_0} (1)\cdot e^{i\omega\tau} \, d\omega$$

$$= \int_{-\omega_0}^{\omega_0} (\cos\omega\tau + i\sin\omega\tau) \, d\omega$$

$$= 2\int_{0}^{\omega_0} \cos\omega\tau \, d\omega .$$

$$= 2\left[\frac{\sin\omega\tau}{\tau}\right]_0^{\omega_0}$$

$$R_{xy}(\tau) = 2\left[\frac{\sin\omega_0\tau}{\tau}\right]$$

W.K.T, $E[x(t)\,x(t+\tau)] = R_{xx}(\tau)$

$$E\left[x(t)\; x\left(t+\frac{\pi}{\omega_0}\right)\right] = R_{xx}\left(\frac{\pi}{\omega_0}\right)$$

$$= \frac{2}{\pi}\,\omega_0 \sin\left(\omega_0 \times \frac{\pi}{\omega_0}\right)$$

$$= \frac{2}{\pi}\,\omega_0 \sin\pi .$$

$[\because \sin\pi = 0]$

$$E\left[x(t)\; x\left(t+\frac{\pi}{\omega_0}\right)\right] = 0$$

$\therefore$ $x(t)$ and $x\left(t+\frac{\pi}{\omega_0}\right)$ are uncorrelated

Cross Spectral Desity :

Consider two jointly wide sense stationary Random process $X(t)$ & $Y(t)$ let $R_{XY}(\tau)$ and $R_{YX}(\tau)$ be their cross correlation functions. Then the cross spectral densities are.

$$S_{XY}(\omega) = \int_{-\infty}^{\infty} R_{XY}(\tau)\, e^{-i\omega\tau}\, d\tau.$$

$$S_{YX}(\omega) = \int_{-\infty}^{\infty} R_{YX}(\tau)\, e^{-i\omega\tau}\, d\tau.$$

(11). The cross power spectrum of real random process $X(t)$ and $Y(t)$ is given by.

i) $S_{XY}(\omega) = \begin{cases} a + \frac{ib\omega}{\alpha}, & -\alpha < \omega < \alpha, \quad \alpha > 0 \\ 0, & \text{otherwise} \end{cases}$

ii) $S_{XY}(\omega) = \begin{cases} a + jb\omega, & |\omega| < 1 \\ 0, & \text{elsewhere} \end{cases}$

Find the cross - correlation function.

Sol:-

i) $$R_{XY}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{XY}(\omega)\, e^{i\omega\tau}\, d\omega.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(a + \frac{ib\omega}{\alpha}\right) e^{i\omega\tau}\, d\omega.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left(\underset{\text{even}}{a} + \underset{\text{odd}}{\frac{ib\omega}{\alpha}}\right) (\underset{\text{even}}{\cos\omega\tau} + \underset{\text{odd}}{i\sin\omega\tau})\, d\omega.$$

$$= \frac{\cancel{2}}{\cancel{2}\pi} \int_{0}^{\infty} \left(a\cos\omega\tau - \frac{b\omega}{\alpha}\sin\omega\tau\right) d\omega.$$

$$= \frac{1}{\pi}\left[a\frac{\sin\omega\tau}{\tau} - \left(\frac{b\omega}{\alpha}\left\{\frac{-\cos\omega\tau}{\tau}\right\} - \frac{b}{\alpha}\left\{\frac{-\sin\omega\tau}{\tau^2}\right\}\right)\right]_{0}^{\infty}.$$

$$= \frac{1}{\pi}\left[a\frac{\sin\alpha\tau}{\tau} + \frac{b\alpha}{\alpha}\left(\frac{\cos\alpha\tau}{\tau}\right) - \frac{b}{\alpha}\left(\frac{\sin\alpha\tau}{\tau^2}\right)\right] - [0+0+0]$$

$$= \frac{1}{\pi}\left[a\frac{\sin\alpha\tau}{\tau} + \frac{b\cos\alpha\tau}{\tau} - \frac{b}{\alpha}\left(\frac{\sin\alpha\tau}{\tau^2}\right)\right]$$

$$= \frac{1}{\pi\alpha\tau^2}\left[a\alpha\tau\sin\alpha\tau + b\alpha\tau\cos\alpha\tau - b\sin\alpha\tau\right]$$

$$R_{XY}(\tau) = \frac{1}{\pi\alpha\tau^2}\left[\sin\alpha\tau\,(a\alpha\tau - b) + b\alpha\tau\cos\alpha\tau\right]$$

ii). $$R_{XY}(\tau) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XY}(\omega)\, e^{i\omega\tau}\, d\omega.$$

$$= \frac{1}{2\pi}\int_{-1}^{1} (a + jb\omega)\, e^{i\omega\tau}\, d\omega.$$

$$= \frac{1}{2\pi}\int_{-1}^{1} (a + jb\omega)(\cos\omega\tau + i\sin\omega\tau)\, d\omega.$$

$$= \frac{2}{2\pi}\int_{0}^{1} (a\cos\omega\tau - b\omega\sin\omega\tau)\, d\omega.$$

$$= \frac{1}{\pi}\left[a\frac{\sin\omega\tau}{\tau} - \left(b\omega\left\{-\frac{\cos\omega\tau}{\tau}\right\} - b\left\{\frac{-\sin\omega\tau}{\tau^2}\right\}\right]_{0}^{1}$$

$$= \frac{1}{\pi}\left[a\frac{\sin\omega\tau}{\tau} + b\omega\frac{\cos\omega\tau}{\tau} - b\frac{\sin\omega\tau}{\tau^2}\right]_0^1$$

$$= \frac{1}{\pi}\left[a\frac{\sin\tau}{\tau} + b\frac{\cos\tau}{\tau} - b\frac{\sin\tau}{\tau^2}\right]$$

$$= \frac{1}{\pi\tau^2}\left[a\tau\sin\tau + b\tau\cos\tau - b\sin\tau\right]$$

$$\therefore R_{xy}(\tau) = \frac{1}{\pi\tau^2}\left[a\tau\sin\tau + b\tau\cos\tau - b\sin\tau\right]$$

Properties of cross power spectrum density.

1) $S_{xy}(\omega) = S_{xy}(-\omega)$

2) $S_{xy}(\omega) = 0$, if $x(t)$ are orthogonal.

3) If $x(t)$ and $y(t)$ are uncorrelated

$$S_{xy}(\omega) = E(x)E(y)\,\delta(\omega)$$

(15) Given that auto correlation function for a stationary ergodic process with no periodic components is $R_{xx}(\tau) = 25 + \frac{4}{1+6\tau^2}$. Find the mean value and variance of the process $x(t)$.

Sol:-

W.K.T, $\mu_x^2 = \lim_{\tau\to\infty} R_{xx}(\tau)$

$$= \lim_{\tau\to\infty}\left[25 + \frac{4}{1+6\tau^2}\right]$$

$$= 25 + \frac{4}{1+6(\infty)^2}$$

$$= 25 + 0$$

$$\mu_x^2 = 25.$$

$$E(x) = \mu_x = 5.$$

$$E[x^2(\tau)] = R_{xx}(0)$$

$$= 25 + \frac{4}{1+6(0)}$$

$$= 25 + 4$$

$$= 29$$

$$Var(x) = E(x^2) - [E(x)]^2$$

$$= 29 - 25$$

$$Var(x) = 4.$$

Linear system:

A system with functional relationship $f(x(t))$ is linear, if for any two input $x_1(t)$ and $x_2(t)$. The output of system can be defined as.

$$f\{a_1x_1(t) + a_2x_2(t)\} = a_1 f\{x_1(t)\} + a_2 f\{x_2(t)\}$$

Time Invariance:

It is defined as a property of linear system that if the input is time shifted by an amount (τ) the corresponding output also time shifted by same amount

If $f(x(t)) = y(t)$ then

$$f(x(t-\tau)) = y(t-\tau), \quad -\infty < \tau < \infty.$$

(13). If $Y(t) = X(t+a) - X(t-a)$. Prove that $R_{YY}(\tau) = 2R_{XX}(\tau) - R_{XX}(\tau+2a) - R_{XX}(\tau-2a)$. Hence Prove that $S_{YY}(\omega) = 4\sin^2 a\omega\, S_{XX}(\omega)$.

Sol:- Given, $Y(t) = X(t+a) - X(t-a)$

W.K.T $R_{XX}(\tau) = E[X(t)\, X(t+\tau)]$

$$E[Y(t)\,Y(t+\tau)] = E[(X(t+a) - X(t-a))\cdot(X(t+a+\tau) - X(t-a+\tau))]$$

$$R_{YY}(\tau) = E[\underbrace{X(t+a)}_{t}\, \underbrace{X(t+a+\tau)}_{t+\tau}] - E[X(t+a)\, X(t-a+\tau)]$$
$$- E[X(t-a)\, X(t+a+\tau)] + E[\underbrace{X(t-a)}_{t}\, \underbrace{X(t-a+\tau)}_{t+\tau}]$$
$$= R_{XX}(\tau) - E[X(t+a)\, X(t+a-2a+\tau)]$$
$$- E[X(t-a)\, X(t-a+\tau+2a)] + R_{XX}(\tau)$$
$$R_{YY}(\tau) = 2R_{XX}(\tau) - R_{XX}(\tau-2a) - R_{XX}(\tau+2a)$$

W.K.T $S_{YY}(\omega) = F[R_{YY}(\tau)]$

$$S_{YY}(\omega) = 2F[R_{XX}(\tau)] - F[R_{XX}(\tau-2a)] - F[R_{XX}(\tau+2a)]$$
$$= 2S_{XX}(\omega) - e^{-i2a\omega} S_{XX}(\omega) - e^{+i2a\omega} S_{XX}(\omega)$$
$$= 2S_{XX}(\omega) - 2S_{XX}(\omega)\left[\frac{e^{i2a\omega} + e^{-i2a\omega}}{2}\right]$$
$$= 2S_{XX}(\omega) - 2S_{XX}(\omega)\cos 2a\omega$$
$$= 2S_{XX}(\omega)\{1 - \cos 2a\omega\}$$
$$= 2S_{XX}(\omega)\, 2\sin^2 a\omega$$
$$= 4\sin^2 a\omega\, S_{XX}(\omega)$$

(14). If the process $X(t)$ is defined as $X(t) = Y(t)Z(t)$, where $Y(t)$ & $Z(t)$ are independent WSS process prove that

i). $R_{XX}(\tau) = R_{YY}(\tau)\, R_{ZZ}(\tau)$ &

ii). $S_{XX}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{YY}(\alpha)\, S_{ZZ}(\omega-\alpha)\, d\alpha.$

Sol:- Given, $X(t) = Y(t)Z(t)$

i) W.K.T $R_{XX}(\tau) = E[X(t)\, X(t+\tau)]$

$$E[X(t)\, X(t+\tau)] = E[Y(t)Z(t)\, Y(t+\tau)Z(t+\tau)]$$
$$R_{XX}(\tau) = E[Y(t)\,Y(t+\tau)]\cdot E[Z(t)\,Z(t+\tau)]$$
$$R_{XX}(\tau) = R_{YY}(\tau)\, R_{ZZ}(\tau).$$

ii). $$S_{XX}(\omega) = \int_{-\infty}^{\infty} R_{XX}(\tau)\, e^{-i\omega\tau}\, d\tau$$

$$F^{-1}[R_{XX}(\tau)] = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{XX}(\omega)\, e^{i\omega\tau}\, d\omega.$$

consider, $$F^{-1}\Big[\overbrace{\int_{-\infty}^{\infty} \{S_{YY}(\alpha)\, S_{ZZ}(\omega-\alpha)\}\, d\alpha}^{S}\Big] = \frac{1}{2\pi}$$

$$\int_{-\infty}^{\infty}\Big(\int_{-\infty}^{\infty} S_{YY}(\alpha)\, S_{ZZ}(\omega-\alpha)\Big) e^{i\omega\alpha}\, d\alpha\, d\omega.$$

let $y = \alpha$ $\quad z = \omega - \alpha \Rightarrow \omega = z + \alpha$

$dy = d\alpha \quad dz = d\omega \quad \omega = z + y$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_{YY}(y) S_{ZZ}(z) e^{i(z+y)\alpha} \, dy\, dz$$

$$= \frac{1}{2\pi} \left\{ \int_{-\infty}^{\infty} S_{YY}(y) e^{iy\alpha} dy \right\} \left\{ \int_{-\infty}^{\infty} S_{ZZ}(z) e^{iz\alpha} dz \right\}$$

$$= \frac{1}{2\pi} \left\{ 2\pi R_{YY}(\alpha) \times 2\pi R_{ZZ}(\alpha) \right\}$$

[let, $\alpha = \tau$]

$$F^{-1}\left[\int_{-\infty}^{\infty} S(\alpha) S_{ZZ}(\omega-\alpha)\, d\alpha \right] = 2\pi \{ R_{YY}(\tau) R_{ZZ}(\tau) \}$$

take fourier on b/s.

$$\frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\alpha) S_{ZZ}(\omega-\alpha)\, d\alpha = F[R_{YY}(\tau) R_{ZZ}(\tau)]$$

$$= F[R_{XX}(\tau)]$$

$$\therefore \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{YY}(\alpha) S_{ZZ}(\omega-\alpha)\, d\alpha = S_{XX}(\omega)$$

Hence proved.

Note:

System in the form of convolution.

$Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du$. If a system is such that its input $x(t)$ and its output $Y(t)$ are related by convolute integral i.e $Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du$ then system is liner time invariant system.

proof:

i) $Y(t)$ linear.

ii) $Y(t)$ time invariant.

Sol:

Given, $Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du$.

W.K.T. $x(t) = a_1 x_1(t) + a_2 x_2(t_2)$

$$Y(t) = \int_{-\infty}^{\infty} h(u) \{ a_1 x_1(t_1-u) + a_2 x_2(t_2-u) \} du$$

$$= a_1 \int_{-\infty}^{\infty} h(u)\, x_1(t_1-u)\, du + a_2 \int_{-\infty}^{\infty} h(u)\, x_2(t_2-u)\, du.$$

$$Y(t) = a_1 Y_1(t_1) + a_2 Y_2(t_2)$$

$\therefore Y(t)$ is linear.

Let, $x(t) = x(t+k)$

$$Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du.$$

$$= \int_{-\infty}^{\infty} h(u)\, x(t+k-u)\, du.$$

$$Y(t) = Y(t+k)$$

$\therefore Y(t)$ is Time invariant.

(16) Show that $X(t)$ is a WSS process then the output $Y(t)$ is a WSS process.

Sol: Given, $X(t)$ is WSS process.

i) $E[X(t)]$ is a constant.

ii) $R(t_1, t_2)$ is fn of $(t_1 - t_2)$

To prove $Y(t)$ is WSS.

i). $E[Y(t)]$ is constant.

ii). $R(t_1, t_2) = E[Y(t_1) Y(t_2)]$ is a fn of $(t_1 - t_2)$.

W.K.T $Y(t) = \int_{-\infty}^{\infty} h(u)\, X(t-u)\, du$.

$$E[Y(t)] = \int_{-\infty}^{\infty} h(u)\, \underbrace{E[X(t-u)]}_{E[X(t)] \text{ is constant}}\, du.$$

$$= E[X(t-u)] \int_{-\infty}^{\infty} h(u)\, du.$$

$\therefore$ $E[Y(t)]$ is a constant.

$$R(t_1, t_2) = E[Y(t_1) Y(t_2)]$$

$$= E\left\{\int_{-\infty}^{\infty} h(u)\, X(t_1-u)\, X(t_2-u)\, du\right\}$$

$$= \int_{-\infty}^{\infty} h(u)\, E[X(t_1-u)\, X(t_2-u)]\, du.$$

$R(t_1, t_2)$ is a fn of $(t_1 - t_2)$

(17) If $\{X(t)\}$ is WSS process and if $Y(t) = \int_{-\infty}^{\infty} h(u)\, X(t-u)\, du$, then

i). $R_{XY}(\tau) = R_{XX}(\tau) * h(-\tau)$ and

ii). $R_{YY}(\tau) = R_{XY}(\tau) * h(\tau)$ $\quad * \rightarrow$ convolution.

iii). $S_{XY}(\omega) = S_{XX}(\omega)\, H^*(\omega)$

iv). $S_{YY}(\omega) = S_{XX}(\omega)\, |H(\omega)|^2$.

Sol: Given, $Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du$.

[$x(t)\, Y(t+\tau) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, x(t+\tau-u)\, du$.

$E[x(t)\, Y(t+\tau)] = \int_{-\infty}^{\infty} E[x(t-u)\, x(t+\tau-u)]\, h(u)\, du$.

$= \int_{-\infty}^{\infty} E[x(\alpha)\, Y(\alpha+\tau)]\, h(t-\alpha)\, (-d\alpha)$

Let $t-u = \alpha$.

$du = -d\alpha$.]

i). $R_{XY}(\tau) = E[X(t)\, Y(t+\tau)]$

$$x(t)\, Y(t+\tau) = x(t) \int_{-\infty}^{\infty} h(u)\, x(t+\tau-u)\, du.$$

$$E[x(t)\, Y(t+\tau)] = \int_{-\infty}^{\infty} h(u)\, E[x(t)\, x(t+\tau-u)]\, du.$$

$$R_{XY}(\tau) = \int_{-\infty}^{\infty} h(u)\, R_{XX}(\tau-u)\, du.$$

Let $u = -\alpha$

$du = -d\alpha$.

$$R_{xy}(\tau) = \int_{+\infty}^{\infty} h(-\alpha)\, R_{xx}(\tau+\alpha)\, (-d\alpha)$$

$$= \int_{-\infty}^{\infty} R_{xx}(\tau+\alpha)\, h(-\alpha)\, d\alpha.$$

$$= \int_{-\infty}^{\infty} h(-\alpha)\, R_{xx}(\tau-(-\alpha))\, d\alpha.$$

$$\therefore R_{xy}(\tau) = R_{xx}(\tau) * h(-\tau) \longrightarrow ①$$

ii). $R_{yy}(\tau) = E[Y(t)\,Y(t+\tau)]$

$$Y(t)\,Y(t+\tau) = Y(t+\tau) \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du.$$

$$E[Y(t)\,Y(t+\tau)] = \int_{-\infty}^{\infty} h(u)\, E[x(t-u)\,Y(t+\tau)]\, du$$

$$R_{yy}(\tau) = \int_{-\infty}^{\infty} R_{xy}(\tau+u)\, h(u)\, du.$$

$$R_{yy}(\tau) = \int_{-\infty}^{\infty} h(u)\, R_{xy}(\tau+u)\, du.$$

$$\therefore R_{yy}(\tau) = R_{xy}(\tau) * h(\tau) \longrightarrow ②.$$

iii). equ ① $\Rightarrow$ $R_{xy}(\tau) = R_{xx}(\tau) * h(-\tau)$

take Fourier transform,

$$F[R_{xy}(\tau)] = F[R_{xx}(\tau) * h(-\tau)]$$

$$S_{xy}(\omega) = S_{xx}(\omega) \cdot F[h(-\tau)]$$

$$[\because F[h(t)] = H(\omega) \quad , \quad F[h(-t)] = H^*(\omega)]$$

$$\therefore S_{xy}(\omega) = S_{xx}(\omega) \cdot H^*(\omega) \longrightarrow ③.$$

iv). equ ② $\Rightarrow$ $R_{yy}(\tau) = R_{xy}(\tau) * h(\tau)$

take fourier transform,

$$F[R_{yy}(\tau)] = F[R_{xy}(\tau) * h(\tau)]$$

$$S_{yy}(\omega) = S_{xy}(\omega) \cdot H(\omega)$$

sub equ ③,

$$S_{yy}(\omega) = [S_{xx}(\omega) \cdot H^*(\omega)]\, H(\omega)$$

$$\therefore S_{yy}(\omega) = S_{xx}(\omega)\, |H(\omega)|^2.$$

(18) If $X(t)$ is a WSS process and if $Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)$ then $R_{yy}(\tau) = R_{xx}(\tau) * k(\tau)$

where, $k(t) = h(t) * h(-t) = \int_{-\infty}^{\infty} h(u)\, h(t+u)\, du.$

Sol: Given, $Y(t) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du.$

$$Y(t)\, Y(t+\tau) = \int_{-\infty}^{\infty} h(u)\, x(t-u)\, du \cdot \int_{-\infty}^{\infty} h(v)\, x(t+\tau-v)\, dv$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(u)\, h(v)\, x(t-u)\, x(t+\tau-v)\, du\, dv.$$

$$E[Y(t)\,Y(t+\tau)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(u)\,h(v)\,E[x(t-u)\,x(t+\tau-v)]\,du\,dv.$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(u)\,h(v)\,R_{xx}(\tau-v+u)\,du\,dv.$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(u)h(v)\,R_{xx}(\tau-(v-u))\,du\,dv.$$

let $\lambda = v-u$

$d\lambda = dv.$

$$= \underbrace{\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} h(u)\,h(\lambda+u)}_{k(t)}\,R_{xx}(\tau-\lambda)\,du\,d\lambda.$$

$$\left[\because k(t) = \int_{-\infty}^{\infty} h(u)\,h(t+u)\,du\right]$$

$$= \int_{-\infty}^{\infty} R_{xx}(\tau-\lambda)\,k(\lambda)\,d\lambda$$

$$\boxed{\therefore R_{yy}(\tau) = R_{xx}(\tau) * k(\tau)}$$

(19). The power spectral densities of the i/p and output processes in the system are connected by the relation $S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$ where $H(\omega)$ is the fourier transform of unit impulse Responce function $h(t)$.

Sol:

W.K.T. $R_{yy}(\tau) = R_{xx}(\tau) * k(\tau)$

take F.T, $F[R_{yy}(\tau)] = F[R_{xx}(\tau) * k(\tau)]$

$$S_{yy}(\omega) = S_{xx}(\omega)\,F[k(\tau)] \rightarrow (1)$$

$$[\because k(\tau) = h(t) * h(-t)]$$

$$F[k(\tau)] = F[h(t) * h(-t)]$$

$$= H(\omega)\,H^{*}(\omega)$$

$$= |H(\omega)|^2$$

$$\boxed{\therefore S_{yy}(\omega) = S_{xx}(\omega)\,|H(\omega)|^2}$$

(20). $x(t)$ is the input voltage to a circuit (system) and $Y(t)$ is the output voltage $x(t)$ is a stationary Random process with $\mu_x = 0$ and $R_{xx}(\tau) = e^{-\alpha|\tau|}$. Find μ_y, $S_{yy}(\omega)$ and $R_{yy}(\tau)$, if power transfer function is $H(\omega) = \frac{R}{R+iL\omega}$.

Sol: Given, $R_{xx}(\tau) = e^{-\alpha|\tau|}$, $\mu_x = 0$.

$$Y(t) = \int_{-\infty}^{\infty} h(u)\,x(t-u)\,du.$$

$$\mu_y = E[Y(t)] = \int_{-\infty}^{\infty} h(u)\,E(x(t-u))\,du.$$

$$\mu_y = 0.$$

W.K.T $S_{YY}(\omega) = S_{XX}(\omega)\,|H(\omega)|^2$

$F[R_{XX}(\tau)] = F[e^{-\alpha|\tau|}]$

$S_{XX}(\omega) = \frac{2\alpha}{\alpha^2+\omega^2}$

$|H(\omega)| = \left|\frac{R}{R+iL\omega}\right| = \frac{R}{\sqrt{R^2+L^2\omega^2}}$

$S_{YY}(\omega) = \left(\frac{2\alpha}{\alpha^2+\omega^2}\right)\left(\frac{R}{\sqrt{R^2+L^2\omega^2}}\right)^2$

$\therefore S_{YY}(\omega) = \frac{2\alpha R^2}{(\alpha^2+\omega^2)(R^2+L^2\omega^2)}$ //

(31). If $X(t)$ is the input voltage to a circuit & $Y(t)$ is the output voltage $\{X(t)\}$ is stationary random process with $\mu_X = 0$ & $R_{XX}(\tau) = e^{-2|\tau|}$. Find mean μ_Y and power spectrum $S_{YY}(\omega)$ of output if system Transfer function is given by $H(\omega) = \frac{1}{\omega+2i}$.

Sol:- Given, $R_{XX}(\tau) = e^{-2|\tau|}$, $\mu_X = 0$.

$Y(t) = \int_{-\infty}^{\infty} h(u)\, X(t-u)\, du$.

$\mu_Y = E[Y(t)] = \int_{-\infty}^{\infty} h(u)\, E[X(t-u)]\, du$

$\therefore \mu_Y = 0$.

W.K.T $S_{YY}(\omega) = S_{XX}(\omega)\,|H(\omega)|^2$.

$F[R_{XX}(\tau)] = F[e^{-2|\tau|}]$

$S_{XX}(\omega) = \frac{2(2)}{2^2+\omega^2} = \frac{4}{4+\omega^2}$.

$|H(\omega)| = \left|\frac{1}{\omega+2i}\right| = \frac{1}{\sqrt{2^2+\omega^2}}$

$S_{YY}(\omega) = \left(\frac{4}{4+\omega^2}\right)\left(\frac{1}{\sqrt{4+\omega^2}}\right)^2$

$S_{YY}(\omega) = \frac{4}{(4+\omega^2)(4+\omega^2)}$

$S_{YY}(\omega) = \frac{4}{(4+\omega^2)^2}$